

Name _____ Teacher _____



GOSFORD HIGH SCHOOL

MATHEMATICS

EXTENSION 1

HSC

2016

ASSESSMENT TASK 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 90 minutes
- Write on one side of your paper only.
- Start a new page for each question.
- Write your name on each page you submit.
- Correct setting out ***must*** be shown or full marks may not be awarded.
- Board approved calculators may be used.
- A Reference sheet is provided.

Total Marks – 54

Section 1 (6 marks)

Questions 1- 6

Section 2 (48 marks)

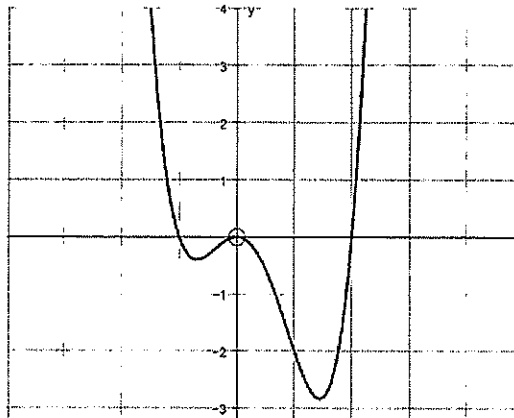
Questions 7 - 9

MULTIPLE CHOICE QUESTIONS TO BE ANSWERED ON SHEET PROVIDED

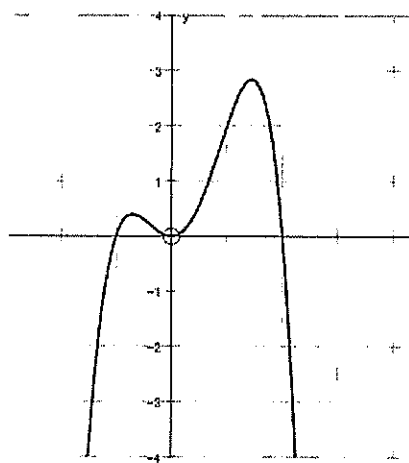
QUESTION 1

Which graph best represents $y = x^4 - x^3 - 2x^2$?

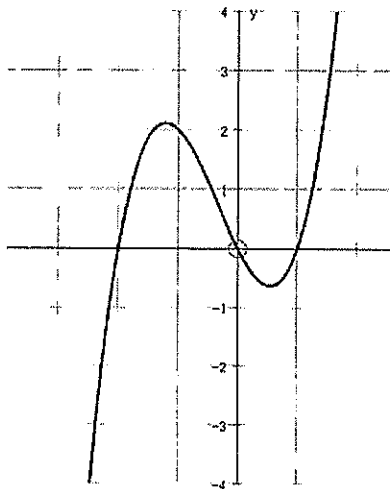
(A)



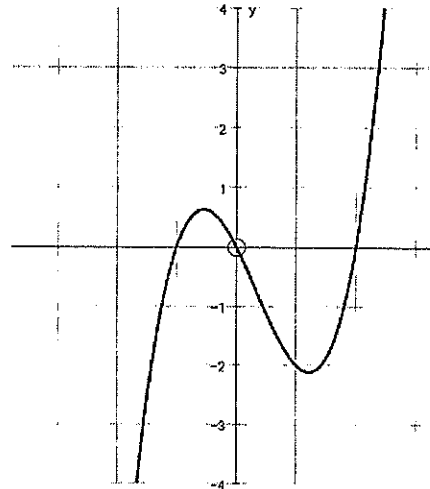
(B)



(C)



(D)



QUESTION 2

A committee of 3 men and 3 women is to be formed from a group of 8 men and 6 women.

In how many ways can this be done?

(A) 48

(B) 1 120

(C) 40 320

(D) 3003

QUESTION 3

Victoria made an error proving that $3^{2n} - 1$ is divisible by 8 using mathematical induction (where n is an integer greater than 0).

Part of the proof is shown below.

Step 2: Assume the result true for $n = k$

$$3^{2k} - 1 = 8P \text{ where } P \text{ is an integer.} \quad \text{Line 1}$$

$$\text{Hence } 3^{2k} = 8P + 1$$

To prove the result is true for $n = k + 1$

$$3^{2(k+1)} - 1 = 8Q \text{ where } Q \text{ is an integer.} \quad \text{Line 2}$$

$$\begin{aligned} \text{LHS} &= 3^{2(k+1)} - 1 \\ &= 3^{2k} \times 3^2 - 1 \\ &= (8P + 1) \times 3^2 - 1 \quad \text{Line 3} \\ &= 72P + 1 - 1 \quad \text{Line 4} \\ &= 72P \\ &= 8(9P) \\ &= 8Q \\ &= \text{RHS} \end{aligned}$$

In which line did Victoria make an error?

- (A) Line 1 (B) Line 2
(C) Line 3 (D) Line 4

QUESTION 4

What is the indefinite integral for $\int (\sin^2 x + x^2) dx$?

- (A) $x - \frac{1}{2} \sin 2x + \frac{x^3}{3} + c$
(B) $\frac{1}{2}x - \frac{1}{4} \sin 2x + \frac{x^3}{3} + c$
(C) $x - \frac{1}{2} \sin 2x + 2x + c$
(D) $\frac{1}{2}x - \frac{1}{4} \sin 2x + 2x + c$

QUESTION 5

Using the substitution $u = 1 - x^3$, evaluate $\int_0^1 x^2 \sqrt{1 - x^3} dx$.

- (A) $-\frac{1}{9}$ (B) $\frac{1}{9}$
(C) $\frac{2}{9}$ (D) $\frac{1}{3}$

QUESTION 6

One approximate solution of the equation $f(x) = 4x^3 - 15x^2 + 22x - 12$ is $x = 1.3$,
What is another approximation to the solution using one of application of Newton's method?

- (A) $x = 1.2884$
- (B) $x = 1.2885$
- (C) $x = 1.2886$
- (D) $x = 1.2887$

QUESTION 7 – (17 MARKS)

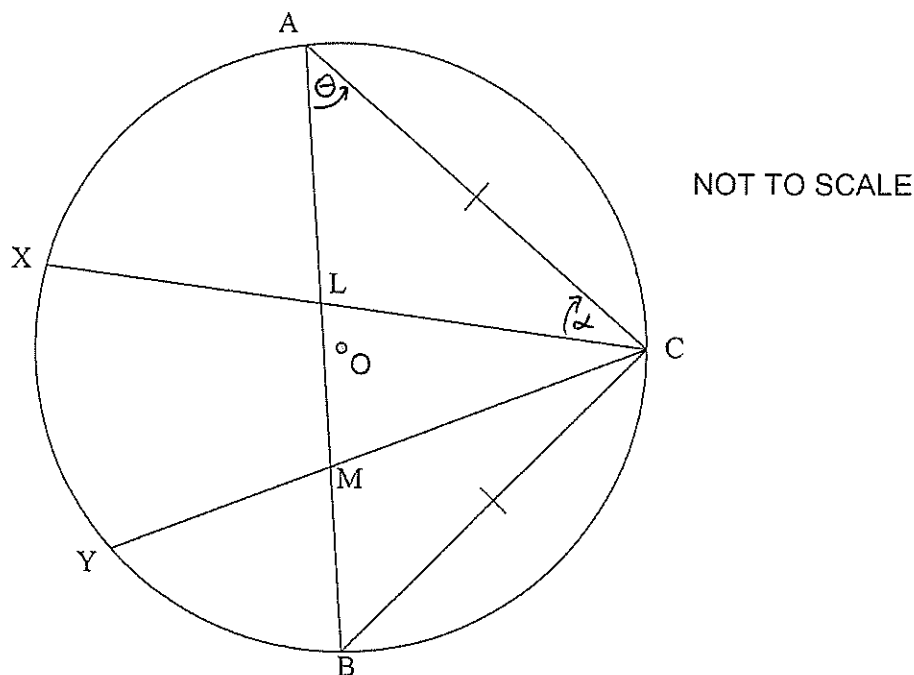
START A NEW PAGE

MARKS

- a) Find the obtuse angle between the lines $2x - 3y + 4 = 0$ and $y = 3x + 7$.
(To the nearest minute)

2

- b) In the diagram A, B and C are 3 points on the circle.
CX and CY are chords cutting AB at L and M respectively.
 $AC = CB$, $\angle CAB = \theta$ and $\angle ACX = \alpha$.
Copy the diagram onto your paper.



- i) State why $\angle CLB = \theta + \alpha$.
- ii) Explain why $\angle AYC = \theta$ and $\angle AYX = \alpha$.
- iii) Prove that XYML is a cyclic Quadrilateral.

1

3

2

QUESTION 7 CONTINUED**MARKS**

c) The point P divides the interval joining $A(-1,4)$ and $B(2,-2)$ in the ratio $k:1$.

i) Write down the coordinates of P in terms of k . 1

ii) Given that P lies on the line $x - 2y - 1 = 0$, find k and hence find the coordinates of P . 3

d) The equation $2x^3 - 6x + 1 = 0$ has roots α , β and γ . Evaluate:

i) $\alpha + \beta + \gamma$ 1

ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1

iii) $(\alpha + 1)(\beta + 1)(\gamma + 1)$ 2

e) $\lim_{x \rightarrow \infty} \left[\frac{x+2}{1-x} \right] = ?$ 1

QUESTION 8 – (15 MARKS) START A NEW PAGE**MARKS**

a) Prove $\frac{2\sin^3 \theta + 2\cos^3 \theta}{\sin \theta + \cos \theta} = 2 - \sin 2\theta$ 3

b) Find the exact value of $\sin \frac{11\pi}{12}$ 3

c) By using the 't' method solve $\sqrt{3} \sin x + \cos x = 1$ for $0 \leq x \leq 2\pi$ 3

d) Evaluate $\int_{3\sqrt{2}}^6 \sqrt{36 - x^2} dx$ using the substitution $x = 6 \sin u$. 4

e) Solve $\frac{x}{x+2} \geq 3$ 2

QUESTION 9 – (16 MARKS)**START A NEW PAGE****MARKS**

- a) i) Show that $P(x) = x^3 - 8x^2 + 9x + 18$ is divisible by $x - 3$ and $x + 1$ 2
- ii) Express $P(x)$ in terms of 3 linear factors. 2
- iii) Hence solve $P(x) \geq 0$ 2
- b) i) Show that $P(x) = 3x^4 + 4x^3 - 12x^2 - 1$ has a root between $x = -3$ and $x = -2$. 1
- ii) Use the method of halving the interval twice to show that the root lies between $x = -3$ and $x = -2.75$ 2
- c) Consider the polynomial $P(x) = 4x^3 + 2x^2 + 1$
 $P(x)$ has a real zero α in the interval $-1 < x < 0$.
- i) By sketching the graph of $P(x)$, show that α is the only real zero of $P(x)$.
(Hint: Show all critical points) 3
- ii) Use Newton's method with initial value $\alpha = -\frac{1}{4}$ to obtain a second approximation for the root. 2
- iii) Explain from the graph of $P(x)$ why this second approximation is *not* a better approximation to α than $\alpha = -\frac{1}{4}$. 1
- iv) Give *one* value of x that would give a better approximation of the root. 1

END OF EXAM



Multiple Choice

1) $y = x^4 - x^3 - 2x^2$

$= x^2(x^2 - x - 2)$

$= x^2(x-2)(x+1)$

double root at $x^2=0 \rightarrow x=0$

single roots at $x-2=0 \rightarrow x=2$

$x+1=0 \rightarrow x=-1$

$a > 0$ (A)

2) ${}^8C_3 \times {}^6C_3 = 1120$ (B)

3) Line 4 should read

$72P + 9 - 1$ (D)

4) $\int (\sin^2 x + x^2) dx$

$\cos 2x = 1 - 2\sin^2 x$

$\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$

$= \frac{1}{2} \int (1 - \cos 2x) dx + \int x^2 dx$

$= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right] + \frac{x^3}{3} + C$

$= \frac{x}{2} - \frac{\sin 2x}{4} + \frac{x^3}{3} + C$ (B)

5) $\int_0^1 x^2 \sqrt{1-x^3} dx$

$u = 1 - x^3$

$\frac{du}{dx} = -3x^2$

$dx = \frac{du}{-3x^2}$

$\frac{1}{3} \int_0^1 u^{1/2} du$

$= \frac{1}{3} \left[\frac{2u^{3/2}}{3} \right]_0^1$

$x^3 = 1 - u$

$x=1 \quad u=0$

$x=0 \quad u=1$

$= \frac{1}{9} (2 - 0)$

$= \frac{2}{9}$ (C)

6) $f(x) = 4x^3 - 15x^2 + 22x - 12$

$f'(x) = 12x^2 - 30x + 22$

$f(1.3) \doteq 0.038$

$f'(1.3) \doteq 3.28$

$x_1 \doteq x_0 - \frac{f(x)}{f'(x)}$

$\doteq 1.3 - \frac{0.038}{3.28}$

$\doteq 1.2884$

(A)

$$7a) 2x - 3y + 4 = 0 \quad m_1 = \frac{2}{3}$$

$$y = 3x + 7 \quad m_2 = 3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad (\text{acute } \angle)$$

$$= \left| \frac{\frac{2}{3} - 3}{1 + \frac{2}{3} \cdot 3} \right| \quad \textcircled{1} \text{ formula}$$

$$= \left| \frac{-\frac{7}{3}}{3} \right|$$

$$= \frac{7}{9}$$

$$\therefore \text{acute } \angle = 37^\circ 52'$$

$$\therefore \text{obtuse } \angle = 142^\circ 8' \quad \textcircled{1}$$

(nearest minute)

2

2i) In $\triangle ACL$

$$\angle LAC = \theta, \angle ACL = \alpha \quad (\text{given})$$

$$\angle CLB = \angle LAC + \angle ACL$$

(Exterior $\angle$ of $\triangle$ = Sum 2 opposite interior $\angle$'s)

$$\therefore \angle CLB = \theta + \alpha$$

1

3) In $\triangle ACB$

$$\angle CAB = \angle CBA = \theta$$

($\angle$'s opposite = sides $\triangle$ =)

$$\therefore \angle CBA = \theta \quad \textcircled{1}$$

$$+ \angle CBA = \angle AXC = \theta$$

($\angle$'s at circumference from same arc =)

$$\angle AXC = \angle ACX = \alpha \quad \textcircled{1}$$

($\angle$'s at circumf from same arc =)

3

iii) for $XYML$

$$\angle XYM = \angle AYC + \angle AYC$$

(adjacent $\angle$'s) ①

$$\therefore \angle XYM = \theta + \alpha$$

$$+ \angle CLB = \theta + \alpha \quad (\text{from above})$$

$$\therefore \angle XYM = \angle CLB$$

$\therefore XYML$ is cyclic quad

(Exterior $\angle$ cyclic quad = opp interior $\angle$) ①

2

ci) $A(-1, 4) \quad B(2, -2)$

$k:1$

$$x = \frac{2k-1}{k+1}$$

$$y = \frac{-2k+4}{k+1}$$

$$\therefore P\left(\frac{2k-1}{k+1}, \frac{-2k+4}{k+1}\right) \quad \textcircled{1}$$

ii) sub P into $x - 2y - 1 = 0$

$$\frac{2k-1}{k+1} - 2\left(\frac{-2k+4}{k+1}\right) - 1 = 0 \quad \textcircled{1}$$

$$\frac{2k-1 + 4k-8-k-1}{k+1} = 0$$

$$\therefore 5k - 10 = 0$$

$$k = 2 \quad \textcircled{1}$$

$$\therefore P(1, 0) \quad \textcircled{1}$$

3

$$d) 2x^3 - 6x + 1 = 0$$

$$i) \sum \alpha = 0$$

1

$$ii) \sum \alpha\beta = \frac{-6}{2} = -3$$

1

$$ii) (\alpha+1)(\beta+1)(\gamma+1)$$

$$= (\alpha\beta + \alpha + \beta + 1)(\gamma + 1)$$

$$= \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \alpha + \beta\gamma + \beta + \gamma + 1$$

$$= \alpha\beta\gamma + \sum \alpha\beta + \sum \alpha + 1 \quad (1)$$

$$= -\frac{1}{2} - 3 + 0 + 1$$

$$= -2\frac{1}{2}$$

2

$$c) \lim_{x \rightarrow \infty} \left(\frac{x+2}{1-x} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{1 + \frac{2}{x}}{\frac{1}{x} - 1} \right)$$

$$= \frac{1}{-1}$$

$$= -1 \quad (\text{must have work})$$

1

Question 8

2) LHS = $\frac{2\sin^3\theta + 2\cos^3\theta}{\sin\theta + \cos\theta}$

$$= \frac{2(\sin\theta + \cos\theta)(\sin^2\theta - \sin\theta\cos\theta + \cos^2\theta)}{\sin\theta + \cos\theta}$$

$$= 2(\sin^2\theta + \cos^2\theta - \sin\theta\cos\theta) \quad (1)$$

$$= 2(1 - \sin\theta\cos\theta) \quad (1)$$

$$= 2 - 2\sin\theta\cos\theta$$

$$= 2 - \sin 2\theta \quad (1)$$

$$= \text{RHS}$$

[3]

2) $\sin \frac{11\pi}{12} = \sin \left(\frac{\pi}{6} + \frac{3\pi}{4} \right) \quad (1)$

$$= \sin \frac{\pi}{6} \cos \frac{3\pi}{4} + \cos \frac{\pi}{6} \sin \frac{3\pi}{4}$$

$$= \frac{1}{2} \cdot -\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \quad (1)$$

$$= \frac{-1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$= \frac{\sqrt{6}-\sqrt{2}}{4} \quad (1)$$

[3]

2) $\sqrt{3}\sin x + \cos x = 1$

$$\sqrt{3} \left(\frac{2t}{1+t^2} \right) + \left(\frac{1-t^2}{1+t^2} \right) = 1 \quad (1)$$

$$\frac{2\sqrt{3}t + 1 - t^2}{1+t^2} = 1$$

$$2\sqrt{3}t + 1 - t^2 = 1 + t^2$$

$$2t^2 - 2\sqrt{3}t = 0$$

$$t(2t - 2\sqrt{3}) = 0$$

$$t = 0 \quad | \quad 2t = 2\sqrt{3}$$

$$t = \sqrt{3}$$

$$\tan \frac{x}{2} = 0 \quad | \quad \tan \frac{x}{2} = \sqrt{3} \quad (1)$$

$$\therefore x = 0, 2\pi, \frac{2\pi}{3} \quad (1)$$

test $x = \pi$

$$\sqrt{3}\sin\pi + \cos\pi = 0 + (-1) \neq 1$$

$\therefore$ not a soln.

d) $\int_{3\sqrt{2}}^6 \sqrt{36-x^2} dx$ $x = 6\sin u$
 $\frac{dx}{du} = 6\cos u$

$$\frac{\pi}{4} \quad \pi/2 \quad dx = 6\cos u \, du$$

$$= \int_{\pi/4}^{\pi/2} (36 - 36\sin^2 u) \cdot 6\cos u \, du \quad x=6 \quad u = \frac{\pi}{2}$$

$$\frac{\pi}{4} \quad (1) \quad x = 3\sqrt{2}, u = \frac{\pi}{4}$$

$$= \int_{\pi/4}^{\pi/2} 6\cos u \cdot 6\cos u \, du \quad (1)$$

$$= 36 \int_{\pi/4}^{\pi/2} \cos^2 u \, du$$

$$\cos 2\theta = 2\cos^2\theta - 1$$

$$\therefore \cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$$

$$= 18 \int_{\pi/4}^{\pi/2} (1 + \cos 2\theta) d\theta \quad (1)$$

$$= 18 \left[\theta + \frac{\sin 2\theta}{2} \right]_{\pi/4}^{\pi/2}$$

$$= 18 \left(\frac{\pi}{2} + 0 - \frac{\pi}{4} - \frac{1}{2} \right)$$

$$= 18 \left(\frac{\pi}{4} - \frac{1}{2} \right) \quad (1)$$

[4]

e) $\frac{x}{x+2} \geq 3$

c.p at

$$x+2=0$$

$$x = -2$$

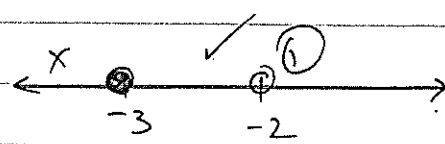
$$x = 3x+6$$

$$2x = -6$$

$$x = -3$$

$$-3 < x < -2$$

[2]



Question 9

2) i) $P(x) = x^3 - 8x^2 + 9x + 18$
 $P(3) = 3^3 - 8(3^2) + 9(3) + 18$
 $= 0$

$\therefore$ by factor theorem $(x-3)$
 is factor (1)

$P(-1) = (-1)^3 - 8(-1)^2 + 9(-1) + 18$
 $= 0$ (1)

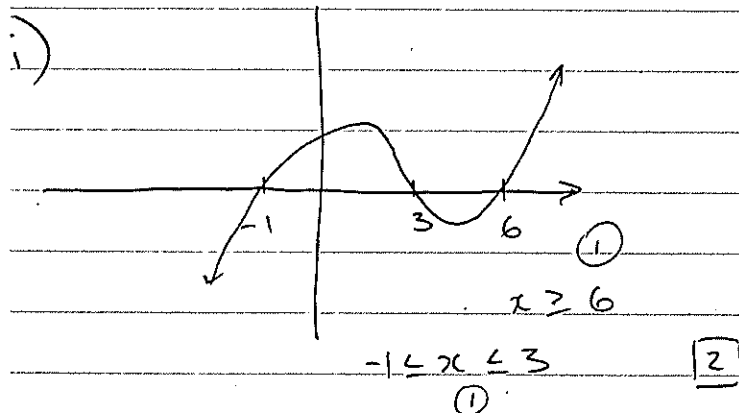
$\therefore (x+1)$ is factor by factor theorem (2)

ii) check $x=6$

$P(6) = 6^3 - 8 \times 6^2 + 9 \times 6 + 18$
 $= 0$ (1)

$\therefore (x-6)$ is a factor

$\therefore P(x) = (x-3)(x+1)(x-6)$ (1) (2)



2) i) $P(x) = 3x^4 + 4x^3 - 12x^2 - 1$
 $P(-3) = 3(-3)^4 + 4(-3)^3 - 12(-3)^2 - 1$
 $= 26 > 0$ (1)
 $P(-2) = 3(-2)^4 + 4(-2)^3 - 12(-2)^2 - 1$
 $= -33 < 0$

$\therefore$ root lies between

$x = -3$ & $x = -2$ (1)

ii) $P(-2.5) = 3(-2.5)^4 + 4(-2.5)^3 - 12(-2.5)^2 - 1$
 $= -21.3125 < 0$

$\therefore$ root lies between

$x = -2.5$ & $x = -3$

$P(-2.75) = 3(-2.75)^4 + 4(-2.75)^3 - 12(-2.75)^2 - 1$
 $= -3.36 < 0$

$\therefore$ root lies between $x = -3$ &

$P(-3) > 0$ $x = -2.75$ (2)

c) $P(x) = 4x^3 + 2x^2 + 1$

i) $P'(x) = 12x^2 + 4x$

$P''(x) = 24x + 4$

stat pts at $P'(x) = 0$

$4x(3x+1) = 0$

$4x = 0$

$x = 0$

$P'' = 4 > 0$

$\therefore$ min turn

$y = 1$

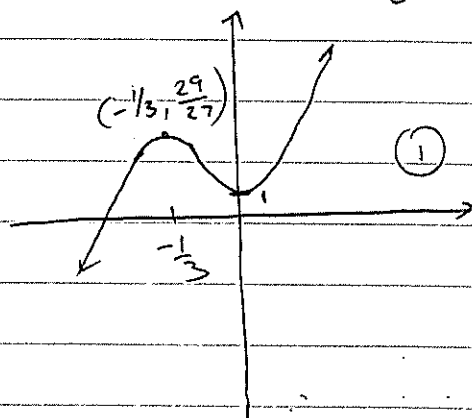
$3x+1 = 0$

$x = -\frac{1}{3}$ (1)

$P'' = - < 0$ (2)

$\therefore$ max turn pt

$y = \frac{29}{27}$



(3)

$$i) P(x) = 4x^3 + 2x + 1$$

$$P'(x) = 12x^2 + 2$$

$$P(-\frac{1}{4}) = \frac{17}{16}$$

$$P'(-\frac{1}{4}) = -\frac{1}{4}$$

①

$$\therefore \alpha_1 = -\frac{1}{4} - \frac{\frac{17}{16}}{-\frac{1}{4}}$$

$$= 4$$

①

2

ii) The tangent at $x = -\frac{1}{4}$
has an x-intercept further
away from the root ①

1

v)

$$x = -\frac{2}{3}$$

(or any other suitable
answer)

1

in the domain

$$-1 < x < -\frac{1}{3}$$